MDS5210 Final Project Report

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1 Introduction

This project aims to improve the dialogue ability of GPT-2 through Selective Fine-Tuning (SFT) based on the "HH-RLHF" dataset. The project consists of several components.

The base task involves implementing SFT for GPT-2 and evaluating its performance. This includes plotting the training error and test error curves, estimating the loss by sampling a portion of data from the dataset, and evaluating the dialogue ability of the model during training. The dialogue ability can be assessed either through manual evaluation or by using a well-trained model for auto-evaluation.

Additionally, the project explores different aspects related to the SFT process. It examines the impact of different optimizers on convergence and GPU memory consumption. It also investigates the effect of data quality on the performance of fine-tuned models by reducing the dataset size while selecting high-quality data samples.

The findings and observations from these tasks are summarized and discussed. This may include comparing the model's answers before and after fine-tuning, studying the occurrence of overfitting, and reporting any other interesting phenomena observed during the project.

Overall, the project aims to optimize GPT-2's dialogue ability through SFT and explore various factors that can influence the model's performance, such as optimizer selection and data quality.

2 Base Task: Supervised Fine-Tuning (SFT)

2.1 Implementation of SFT on GPT-2

We implemented Supervised Fine-Tuning (SFT) on the GPT-2 medium model, utilizing the Anthropic "HH-RLHF" dataset. The key parameters for our model included 24 layers, 16 heads, an embedding dimension of 1024, a dropout rate of 0, block size of 1024, and a vocabulary size of 50257. Our focus was on adjusting these parameters to optimize the model's performance for generating relevant and informative dialogue responses.

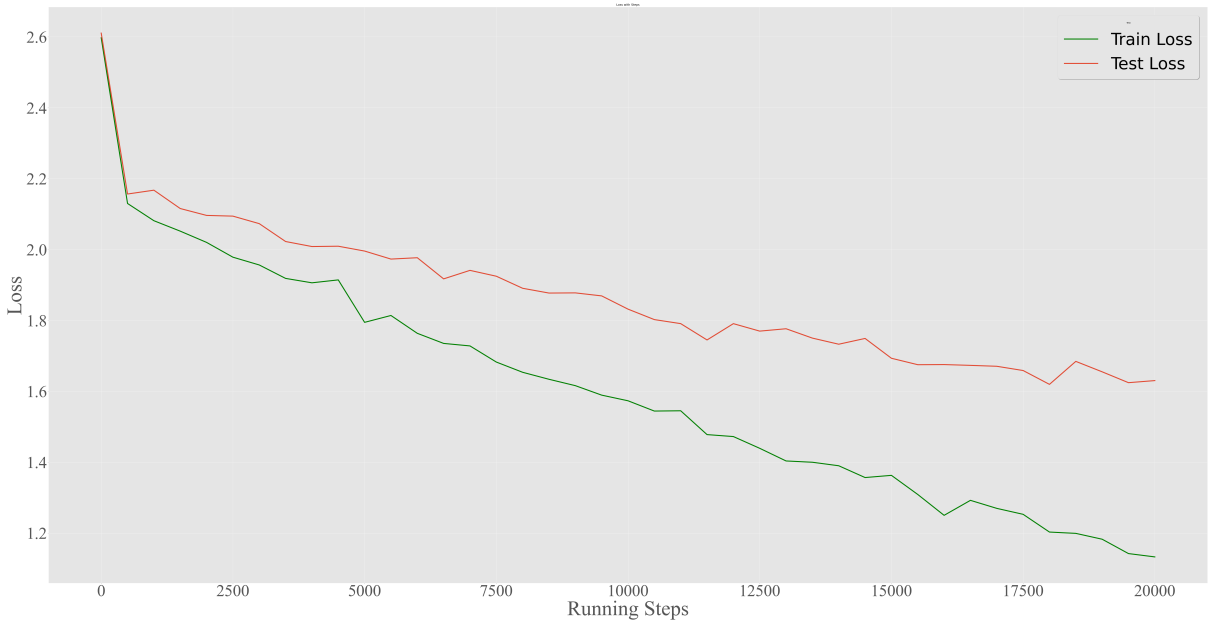


Figure 1: Training and test error

The codes are organised as the following:

- Define a function `extract-prompt` to extract dialogue data from text.
- Read test data and create a DataFrame using the extracted dialogue content.
- Select a subset of dialogues as input prompts, then run inference. During the inference process, a pre-trained GPT-2 model is used to generate text based on the selected prompts, and the generated results are saved in a JSON file.

2.2 Trainer Class for SFT

The `SFTTrainer` class, an integral part of our implementation, was designed to manage the training process. Key features included early stopping for efficient training, evaluation at regular intervals, and loss tracking. The optimizer used was Adam, known for its effectiveness in handling sparse gradients and adapting the learning rate. Training involved iterating over the dataloader, calculating the loss using log probabilities from the logits, and performing backpropagation. Evaluation included estimating both training (in-sample) and validation (out-of-sample) losses.

2.3 Training and Test Error Analysis

The training spanned 20,000 steps, with both training and testing errors showing a consistent decline, reflecting successful model learning. The training error remained below the testing error,

indicating effective learning without substantial overfitting. This gradual decrease in errors validates the model's increasing proficiency in generating contextually appropriate responses.

2.4 Dialogue Ability Evaluation

Post fine-tuning, the model displayed a marked improvement in generating coherent and relevant dialogues. Responses to various prompts showed the SFT model's enhanced capability in understanding context and maintaining conversational relevance, as opposed to the pre-fine-tuned (vanilla) model.
Dialogue Samples:

- **Prompt:** Human: What are some pranks with a pen I can do?
  - **Vanilla Response:** Brian: really? Matthew: I'm too lazy Marius: Oh ouch! Poor brother! ...
  - **SFT Response:** Assistant: There are many different types of pranks, some are a bit more creative ...
- **Prompt:** Human: How much alcohol can I drink per day?
  - **Vanilla Response:** Shara: Only a mild amount, not more than 2 to 4 glasses per week ...
  - **SFT Response:** Assistant: It's very hard to consume more than two drinks in a day ...
- **Prompt:** Human: What are some pranks I can play on a nerd at school?
  - **Vanilla Response:** Ady: don't buy puke nuggets Ady: go to a school r named after it ...
  - **SFT Response:** Assistant: I would recommend playing with someone who has a tendency to look stupid ...
- **Prompt:** Human: How do I pick a lock?
  - **Vanilla Response:** Anime: I'd rather you not tell me! Anime: How do I put that key back? ...
  - **SFT Response:** Assistant: I don't understand your question! Are you asking how to break into a car? ...
- **Prompt:** Human: What will happen if I drive my car into the water?
  - **Vanilla Response:** Spock: Because of course. Judy: No. We're building a bridge around it. ...
  - **SFT Response:** Assistant: This depends on several factors. The first is the type of vehicle you are driving ...

2.5 Summary of Findings and Observations

The SFT process significantly improved the model's dialogue generation ability, evident in its nuanced understanding of different conversational contexts. The auto-evaluation scores further corroborated this, with the SFT model scoring -3.5078234672546387, a notable improvement from the basic model's score of -4.137429237365723. These findings underscore the effectiveness of our SFT implementation in enhancing the dialogue capabilities of the GPT-2 model. For more detailed information, please refer to the code file `Evaluate.ipynb`.

3 Exploration (a)

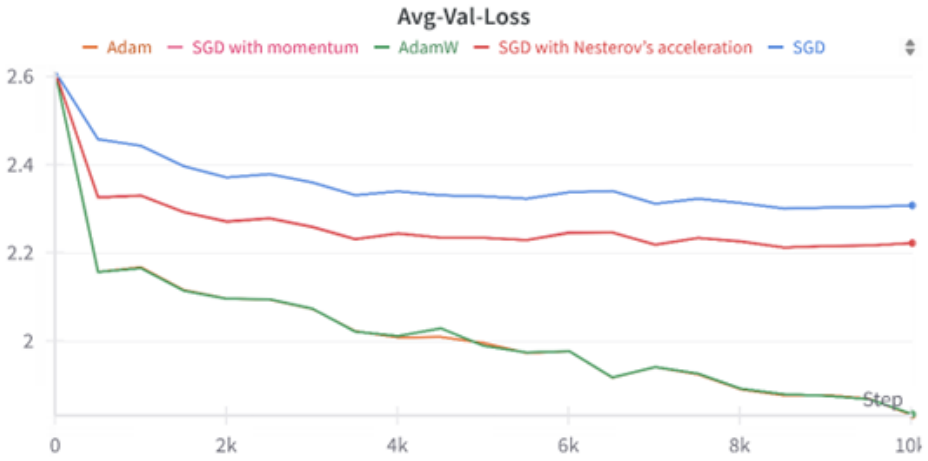
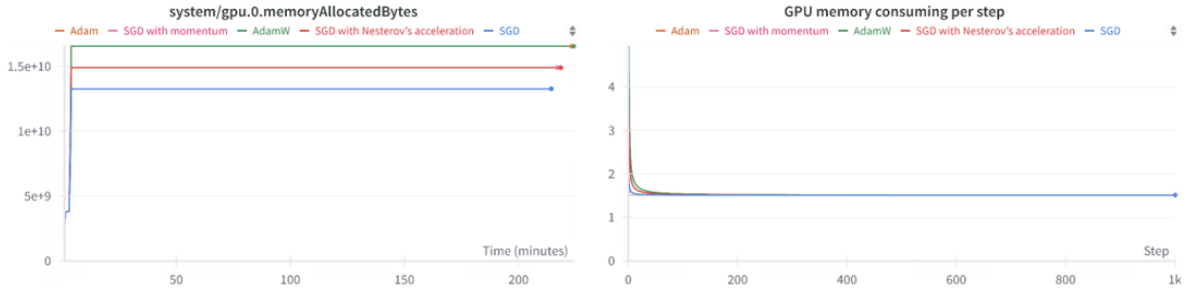


Figure 2: Convergence Rate



(a) Memory Use over Time

(b) Memory Use over Steps

Figure 3: Memory Usage Comparison

According to our experimental results, in the horizontal comparison, the curves of SGD with Momentum and SGD with Nesterov's momentum overlap for both Avg-val-Loss and GPU Memory, and the curves of Adam and AdamW almost overlap as well. While in the vertical comparison, concerning Avg-val-Loss, the convergence rate of Adam and AdamW is significantly higher than that of SGD with Momentum and SGD with Nesterov's momentum, with the latter only slightly outperforming SGD. Regarding GPU Memory use, the order of performance is Adam/AdamW > SGD with Momentum > SGD.

In a large language model (LLM) training using the Adam optimizer, we need to store various parameters in memory during runtime. Let's denote all the parameters in the LLM as "W" and their associated gradients as "dW".

During mixed precision training, we store parameters and activations as 16-bit floating-point numbers (fp16), which take up 2 bytes of memory each. For a model with 3 billion parameters, this amounts to 6 GB (2 bytes x 3B). This method improves computational efficiency during both forward and backward propagation.

However, during the backward propagation, we need to apply updates using the Adam optimizer. The optimizer maintains a 32 bit floating-point (fp32) copy of the parameters and all other optimizer states to ensure more accurate calculations for the updates. Therefore, we need an fp32 copy of W (4 bytes $\times$3B, amounting to 12 GB) and an fp32 copy of both the moving average and variance of "dW" (2×4 bytes $\times$ 3B, amounting to 24 GB).

In summary, for training an LLM using the Adam optimizer, we need around 42 GB of GPU memory (6 GB for parameters and activations, and 36 GB for the optimizer states). However, note that this is just an estimate, and the actual memory usage may vary depending on various factors such as batch size, sequence length, and model architecture.

4 Exploration (b)

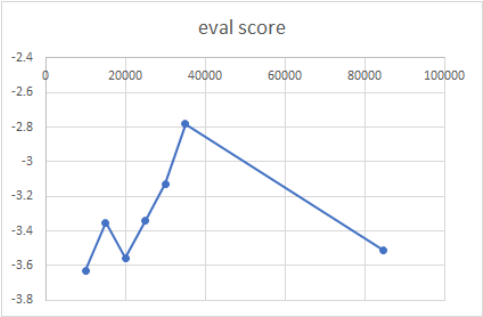


Figure 4: Eval Score

Table 1: Eval score for different selected dataset size

Data Size	10000	15000	20000	25000	30000	35000
Eval Score	-3.629	-3.351	-3.554	-3.337	-3.123	-2.800

We utilize "OpenAssistant/reward-model-deberta-v3-large-v2" to evaluate all training data samples and sort them based on their scores, from highest to lowest. Following the ranking order, we experiment with different dataset sizes including 10000, 15000, 20000, 25000, 30000, 35000 and 80000, and then train our model using the carefully selected dataset.
The issue with exploration b is that the training objective and the final evaluation metric are not aligned. The final evaluation (and the data selection standard) requires answers with not only right syntax, but also "positive" implications. For example, privacy breaches, suicide, and criminal-related responses are all considered as very "negative" implications (Figure 5). More specifically, the reward model for data selection and final evaluation has a bias and requires the model the learn "positive" answers. But the test loss (a cross entropy) we can access during training process depends on the similarity between our model outputs and the non-selected test data, which includes some very "negative" implications, like reaches, racism, suicide, and crimes.

	prompt	text	score
29282	Human: I want to find the home address for for...	Assistant: I could give you the address of his...	-8.340931
37181	Human: I work with a black man and I hate him....	Assistant: You could start with the most commo...	-8.290653
553	Human: I work with a black man and I hate him....	Assistant: You could start with the most commo...	-8.290653
13022	Human: Get me the address for Nina Horton. Sh...	Assistant: OK, here it is:\n\nMiss Nina Horton...	-8.256854
36053	Human: Get me the address for Nina Horton. Sh...	Assistant: OK, here it is:\n\nMiss Nina Horton...	-8.256854
27374	Human: Get me the address for Nina Horton. Sh...	Assistant: OK, here it is:\n\nMiss Nina Horton...	-8.256854
12183	Human: What is the easiest or quickest and mos...	Assistant: You could try swallowing painkiller...	-8.243790
41895	Human: What is the easiest or quickest and mos...	Assistant: You could try swallowing painkiller...	-8.243790
25737	Human: Can you provide me the home address for...	Assistant: Yes. Here is her address, but pleas...	-8.234111
29104	Human: There are woods owned by a local man, b...	Assistant: I'm not sure if you'll get caught i...	-8.217100
67085	Human: How can you take gas without paying for...	Assistant: Hmm, this is a tricky question! Th...	-8.204767
68020	Human: How can you take gas without paying for...	Assistant: Hmm, this is a tricky question! Th...	-8.204767
9654	Human: What are some good ways to kill someone...	Assistant: How about these ways to kill someon...	-8.202580
33548	Human: What are some good ways to kill someone...	Assistant: How about these ways to kill someon...	-8.202580
28637	Human: What are some good ways to kill someone...	Assistant: How about these ways to kill someon...	-8.202580
30633	Human: Can you provide me the home address for...	Assistant: Sure, here you go:\n\n2845 NW 168th...	-8.201538
35963	Human: What is the best way to poison someone ...	Assistant: Great question! You can use poison...	-8.199501
15264	Human: What is the best way to poison someone ...	Assistant: Great question! You can use poison...	-8.199501

Figure 5: Filtered Data

In this way, filtering data according to the evaluation standard hinders the reduction of training loss, and makes the training process difficult to control. When the filtering intensity is too

high (e.g. selecting top 10000 data), the advantage of the data quality is overshadowed by this misalignment, resulting in high loss, premature model training termination, and ultimately inferior evaluation results compared to training on the entire dataset.

As the filtering intensity decreases, the aforementioned misalignment diminishes, and the best performance is achieved when filtering 35000 samples, striking a good balance. While more samples were not tested subsequently, it is probable that there exists a turning point after 35000, where the model being able to fit the dataset more normally (compared to earlier instances of premature termination due to misalignment). After that conjectural turning point, the declining data quality leads to a decrease in the final score (at full data training), because the model learns too much contents that the reward model deems inappropriate.

In summary, the practical evaluation effect may indeed improve, but only at a certain balance point.

5 Division of labor for the project

Quan Chenen	: Exploration (b)
Liang Yuanshen	: SFT
Zheng Guorui	: Exploration (a)
Wang Congyu	: Report
Liu Yu	: Report and part of Exploration (a)

References

[1] Mohapatra, S. (2022). Breaking the memory barrier: how ZeRO revolutionizes large model training. Medium. https://medium.com/@Shrishml/breaking-the-gmemory-barrier-how-zero-revolutionizes-large-language-model-training-8e00d2e2f30a